



Student Number:

SCEGGS Darlinghurst

2007

Higher School Certificate
Assessment Task 1

Mathematics-Extension I

Task Weighting: 25%

Outcomes Assessed: PE2, PE3, HE2, HE7

General Instructions

- Time allowed – 65 minutes
- Start each question on a new page.
- Attempt **all** questions and show all necessary working.
- Write your student number at the top of each page.
- Marks can be deducted for careless or badly arranged work
- Mathematical templates, geometrical equipment and scientific calculators may be used.

Question	Reasoning	Communication	Total
1	/4		/10
2	/3	/3	/12
3	/8	/4	/12
4	/7	/5	/12
Total	/22	/12	/46

Average: _____

St. Dev.: _____

Rank: _____

Parent's Signature _____

START A NEW PAGE

Question 1 (10 Marks)

Marks

- a) Find the general solution, in terms of π , to:

2

$$2\sin x = \sqrt{3}$$

- b) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$

1

- c) i) Show that $(x + 2)$ is a factor of the polynomial:

2

$$P(x) = 4x^3 + x^2 - 11x + 6$$

and hence solve $P(x) = 0$ completely.

- ii) Hence solve $4\cos^3 \theta + \cos^2 \theta = 11\cos \theta - 6$ for $0 \leq \theta \leq 2\pi$

2

Give answers to 2 decimal places.

- d) i) How many eleven-letter arrangements can be made using the letters of the word:

1

YARRAWARRAH

- ii) What is the probability that an arrangement of the letters, chosen at random, has all the A's next to each other?

2

START A NEW PAGE

Question 2 (12 Marks)

Marks

- a) Given α, β, γ are the roots of the equation $3x^3 - 4x^2 + 7x - 5 = 0$.

Write down the values of:

i) $\alpha + \beta + \gamma$ 1

ii) $\alpha\beta + \beta\gamma + \gamma\alpha$ 1

iii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 2

iv) $(\alpha - 1)(\beta - 1)(\gamma - 1)$ 2

- b) Find the roots of $4x^3 - 13x^2 - 13x + 4 = 0$ given that they are the first three terms of a geometric series. 3

- c) Prove by Mathematical Induction that: 3

$$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \frac{1}{9 \times 13} + \dots + \frac{1}{(4n-3)(4n+1)} = \frac{n}{4n+1} \quad \text{for } n \geq 1$$

START A NEW PAGE

Question 3 (12 Marks)

Marks

- a) The polynomial $P(x) = x^3 - x^2 - 2x - 3$ has the same remainder when divided by $(x + a)$ and $(x - 2a)$.
Find the non-zero values of a . 2
- b) i) Express $\tan 2A$ in terms of $\tan A$ 1
- ii) Hence, or otherwise, show that $\tan \frac{\pi}{8} = \sqrt{2} - 1$ 2
- c) There are 18 students in a kindergarten class. Each week the teacher appoints three students to be a lunch monitor, class captain and recycling collector.
- i) How many different appointments can the teacher make to the three positions in the first week? 2
- ii) Erin and Amy are both students in the class. What is the probability that both of them are appointed to any of the three positions at this time? 2
- d) Prove by mathematical induction that $2 \times 4^{2n+1} + 3^{3n+1}$ is divisible by 11 for all positive integers $n \geq 1$ 3

START A NEW PAGE

Question 4 (12 Marks)

Marks

- a) A standard pack of 52 cards consists of 13 cards of each of the four suits: spades, hearts, clubs and spades.
- i) In how many ways can seven cards be selected without replacement so that exactly 4 are diamonds and 3 are hearts? **2**
(Note: The order of the selection is not important)
- ii) In how many ways can seven cards be selected without replacement if at least five must be of the same suit? **2**
(Note: The order of the selection is not important)
- iii) Explain the calculations in your solution for part ii) **2**
- b) i) Express $5\cos x - 3\sin x$ in the form $R\cos(x + a)$ **1**
- ii) Hence find the minimum value of $5\cos x - 3\sin x$ and the value of x , in the domain $0 \leq x \leq \pi$, where this minimum occurs. **3**
(Give your answer to 2 decimal places)
- iii) Graph $y = 5\cos x - 3\sin x$ for $0 \leq x \leq \pi$. Label the x and y intercepts, the turning point and the end points. **2**

END OF TEST

2007 HSC Assessment Task 1 - Extension 1:

Q1. a) $2 \sin x = \sqrt{3}$

$$\sin x = \frac{\sqrt{3}}{2} \quad \checkmark$$

$$x = n\pi + (-1)^n \left(\frac{\pi}{3}\right) \quad \checkmark$$

b) $\lim_{n \rightarrow \infty} \frac{\sin 3n}{2n}$

$$= \frac{1}{2} \lim_{n \rightarrow \infty} \frac{\sin 3n}{n}$$

$$= \frac{3}{2} \lim_{n \rightarrow \infty} \frac{\sin 3n}{3n}$$

$$= \frac{3}{2} \times 1 \quad \text{as } \lim_{n \rightarrow \infty} \frac{\sin an}{an} = 1$$

$$= \frac{3}{2} \quad \checkmark$$

c) i) $P(-2) = 4(-2)^3 + (-2)^2 - 11(-2) + 6$

$$= -32 + 4 + 22 + 6 \quad \checkmark$$

$$= 0$$

$\therefore$ by the factor theorem

as $P(-2) = 0$ $(x+2)$ is a factor

$$\begin{array}{r} 4x^2 - 7x + 3 \\ x+2 \overline{) 4x^3 + x^2 - 11x + 6} \\ \underline{4x^3 + 8x^2} \\ -7x^2 - 11x + 6 \\ \underline{-7x^2 - 14x} \\ 3x + 6 \\ \underline{3x + 6} \\ 0 \end{array}$$

$$\therefore P(x) = (x+2)(4x^2 - 7x + 3)$$

$$= (x+2)(4x-3)(x-1)$$

$$\therefore P(x) = 0$$

$$(x+2)(4x-3)(x-1) = 0$$

$$x = -2 \quad x = \frac{3}{4} \quad x = 1 \quad \checkmark$$

$$c) ii) 4\cos^3\theta + \cos^2\theta = 11\cos\theta - 6$$

$$\text{let } x = \cos\theta$$

$$4x^3 + x^2 = 11x - 6$$

$$4x^3 + x^2 - 11x + 6 = 0$$

$$(x+2)(x-1)(4x-3) = 0 \text{ from i)}$$

$$\therefore \cos\theta = -2 \quad \cos\theta = 1 \quad \cos\theta = \frac{3}{4} \quad \checkmark$$

$$\text{no soln} \quad \theta = 0, 2\pi \quad \theta = 0.72$$

$$\theta = 2\pi - 0.72$$

$$= 5.56$$

$$\therefore \theta = 0, 0.72, 5.56, 6.28. \quad \checkmark \quad \text{Reason-2}$$

$$d) i) \text{ No of arrangements} = \frac{11!}{4!4!}$$

$$= 69300 \quad \checkmark$$

$$\text{No of arrangements with AAAA} = 8 \times \frac{7!}{4!} \quad \checkmark$$

$$= 1680$$

$$P(\text{AAAA}) = \frac{1680}{69300}$$

$$= \frac{4}{165} \quad \checkmark \quad \text{Reason-2}$$

$$\begin{aligned} \underline{Q2} \text{ a) i) } \alpha + \beta + \gamma &= \frac{-5}{a} \\ &= \frac{-(-4)}{-3} \\ &= \frac{4}{3} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{ii) } \alpha\beta + \beta\gamma + \alpha\gamma &= \frac{c}{a} \\ &= \frac{7}{3} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{iii) } \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\beta\gamma}{\alpha\beta\gamma} + \frac{\alpha\gamma}{\alpha\beta\gamma} + \frac{\alpha\beta}{\alpha\beta\gamma} \\ &= \frac{\alpha\beta + \beta\gamma + \alpha\gamma}{\alpha\beta\gamma} \quad \checkmark \\ &= \frac{\frac{7}{3}}{\frac{4}{3}} = \frac{7}{4} \quad \checkmark \end{aligned}$$

$$\begin{aligned}
 \text{a) i) } & (\alpha-1)(\beta-1)(\gamma-1) \\
 & (\alpha\beta - \alpha - \beta + 1)(\gamma-1) \\
 & \alpha\beta\gamma - \alpha\beta - \alpha\gamma - \beta\gamma + \gamma + \alpha + \beta - 1 \checkmark \\
 & = \frac{5}{3} - \frac{7}{3} + \frac{4}{3} - 1 \\
 & = -\frac{1}{3} \checkmark
 \end{aligned}$$

b) let the roots be $\frac{a}{r}, a, ar$

$$\begin{aligned}
 \frac{a}{r} \times a \times ar &= -\frac{4}{4} \\
 a^3 &= -1 \\
 a &= -1 \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \frac{a}{r} + a + ar &= \frac{13}{4} \\
 -\frac{1}{r} - 1 - r &= \frac{13}{4} \\
 -1 - r - r^2 &= \frac{13r}{4} \\
 -4 - 4r - 4r^2 &= 13r \\
 4r^2 + 17r + 4 &= 0 \checkmark \\
 (4r+1)(r+4) &= 0 \\
 r = -\frac{1}{4} \quad r = -4 \\
 \therefore \text{ roots are } 4, -1, \frac{1}{4} \checkmark \quad \text{Reqs} = 3
 \end{aligned}$$

c) Step 1: Show result is true for $n=1$

$$\begin{aligned}
 \text{LHS} &= \frac{1}{1 \times 5} = \frac{1}{5} & \text{RHS} &= \frac{1}{4 \times 1 + 1} \\
 & & &= \frac{1}{5}
 \end{aligned}$$

$\therefore \text{LHS} = \text{RHS} \checkmark$

$\therefore$ the result is true for $n=1$

Step 2: Assume the result is true for $n=k$

$$\text{i.e. } \frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \dots + \frac{1}{(4k-3)(4k+1)} = \frac{k}{4k+1}$$

Step 3: Show the result is true for $n=k+1$

$$\text{i.e. } \frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \dots + \frac{1}{(4k-3)(4k+1)} + \frac{1}{(4k+1)(4k+5)} = \frac{k+1}{4k+5}$$

$$\text{LHS} = \frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \dots + \frac{1}{(4k-3)(4k+1)} + \frac{1}{(4k+1)(4k+5)}$$

$$= \frac{k}{4k+1} + \frac{1}{(4k+1)(4k+5)} \quad (\text{from step 2}) \quad \checkmark$$

$$= \frac{k(4k+5) + 1}{(4k+1)(4k+5)}$$

$$= \frac{4k^2 + 5k + 1}{(4k+1)(4k+5)}$$

$$= \frac{(4k+1)(k+1)}{(4k+1)(4k+5)} \quad \checkmark$$

$$= \frac{k+1}{4k+5} \quad \text{Comm. -3}$$

= RHS

$\therefore$ the result is true for $n=k+1$

$\therefore$ by the Principle of Induction the result is true for $n \geq 1$

$$\text{Q3 a) } P(-a) = (-a)^3 - (-a)^2 - 2(-a) - 3 \\ = -a^3 - a^2 + 2a - 3$$

$$P(2a) = (2a)^3 - (2a)^2 - 2(2a) - 3 \\ = 8a^3 - 4a^2 - 4a - 3$$

$$\therefore \text{ if } P(-a) = P(2a)$$

$$\text{then } -a^3 - a^2 + 2a - 3 = 8a^3 - 4a^2 - 4a - 3 \quad \checkmark$$

$$0 = 9a^3 - 3a^2 - 6a$$

$$0 = 3a(3a^2 - a - 2)$$

$$0 = 3a(3a+2)(a-1)$$

$\therefore$ non-zero values of a are

$$a=1 \quad \text{and} \quad a = -\frac{2}{3} \quad \checkmark \quad \text{Ans-2}$$

$$b) i) \tan(A+A) = \frac{\tan A + \tan A}{1 - \tan A \tan A}$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A} \quad \checkmark \quad \text{Comm.-1}$$

$$ii) \text{ let } A = \frac{\pi}{8}$$

$$\therefore \tan \frac{\pi}{4} = \frac{2 \tan \frac{\pi}{8}}{1 - \tan^2 \frac{\pi}{8}}$$

$$1 = \frac{2 \tan \frac{\pi}{8}}{1 - \tan^2 \frac{\pi}{8}} \quad \checkmark$$

$$1 - \tan^2 \frac{\pi}{8} = 2 \tan \frac{\pi}{8}$$

$$\tan^2 \frac{\pi}{8} + 2 \tan \frac{\pi}{8} - 1 = 0$$

$$\text{let } t = \tan \frac{\pi}{8}$$

$$t^2 + 2t - 1 = 0$$

$$t = \frac{-2 \pm \sqrt{2^2 - 4 \times 1 \times (-1)}}{2 \times 1}$$

$$= \frac{-2 \pm \sqrt{8}}{2}$$

$$= \frac{-2 \pm 2\sqrt{2}}{2}$$

$$= -1 \pm \sqrt{2}$$

since $\frac{\pi}{8}$ is in the first quadrant

$$\tan \frac{\pi}{8} > 0$$

$$\therefore \tan \frac{\pi}{8} = \sqrt{2} - 1 \quad \checkmark \quad \text{Reas-2}$$

$$c) i) \text{ No. of appointments} = {}^{18}C_3 \times 3! \quad \checkmark$$

$$= 4896$$

$\checkmark$ Reas-2

$$ii) \text{ No. of arrangements with Ern \& Amy}$$

$$= {}^{16}C_1 \times 1 \times 1 \times 3!$$

$$= 96 \quad \checkmark$$

$$\therefore P(\text{Ern \& Amy}) = \frac{96}{4896}$$

$$= \frac{1}{51}$$

$\checkmark$ Reas-2

d) Step 1: show true for $n=1$

for $n=1$

$$2 \times 4^3 + 3^4$$

$$= 2 \times 64 + 81$$

$$= 128 + 81$$

$$= 209 \quad \checkmark$$

$$= 11 \times 19$$

$\therefore$ the result is true for $n=1$

Step 2: assume the result is true for $n=k$

$$\text{i.e. } 2 \times 4^{2k+1} + 3^{3k+1} = 11M$$

where M is any integer

Step 3: show the result is true for $n=k+1$

$$\text{i.e. } 2 \times 4^{2k+3} + 3^{3k+4} = 11Q$$

where Q is any integer

$$\text{LHS} = 2 \times 4^{2k+3} + 3^{3k+4}$$

$$= 4^2 (2 \times 4^{2k+1}) + 3^3 (3^{3k+1})$$

$$= 16 (11M - 3^{3k+1}) + 27 (3^{3k+1}) \quad (\text{from step 2}) \quad \checkmark$$

$$= 176M - 16 \times 3^{3k+1} + 27 \times 3^{3k+1}$$

$$= 176M + 11 \times 3^{3k+1}$$

$$= 11 (16M + 3^{3k+1}) \quad \checkmark$$

$$= 11Q \quad \text{where } Q \text{ is any integer}$$

$\Rightarrow M$ and k are integers

$\therefore$ by the Principle of Induction the result is true for $n \geq 1$ Comm-3

Q4 a) i) No. of ways = ${}^{13}C_4 \times {}^{13}C_3$ ✓
 $= 204490$ ✓ Recs - 2

ii) 5 the same: No. of ways = ${}^{13}C_5 \times {}^{39}C_2$
 $= 953667$

6 the same: No. of ways = ${}^{13}C_6 \times {}^{39}C_1$
 $= 66924$

7 the same: No. of ways = ${}^{13}C_7$
 $= 1716$

$\therefore \text{total} = 1022307$

$\therefore \text{total no of ways} = 1022307 \times 4$ ✓
 $= 4089228$ Recs - 2

iii) "at least 5 the same" means 5, 6, or 7 the same

• from 1 suits of 13 choose either

5, 6 or 7 and the remainders (2, 1 or 0)

from 39 cards

• had to multiply by 4 for the 4

different suits

any 2 of them ✓
 Com - 2

b) i) $5 \cos x - 3 \sin x = R \cos x \cos a - R \sin x \sin a$

$\therefore R \cos a = 5 \dots \textcircled{1}$

$R \sin a = 3 \dots \textcircled{2}$

$\textcircled{2} \div \textcircled{1} \quad \tan a = \frac{3}{5}$

$a = 0.54$

$R^2 (\cos^2 a + \sin^2 a) = 5^2 + 3^2$

$R^2 (\cos^2 a + \sin^2 a) = 25 + 9$

$R^2 = 34$

✓ Com - 1

$R = \sqrt{34} \quad \therefore 5 \cos x - 3 \sin x = \sqrt{34} \cos(x + 0.54)$

ii) min value = $-\sqrt{34}$ ✓

$\therefore \sqrt{34} \cos(x + 0.54) = -\sqrt{34}$

$\cos(x + 0.54) = -1$ ✓

$x + 0.54 = \pi$

Recs - 3

$x = 2.60$ (to 2 d.p.) ✓

iii) sketch $y = 5 \cos x - 3 \sin x$

$$\therefore \text{sketch } y = \sqrt{34} \cos(x + 0.54)$$

turning point = min turning point

$$\therefore (2.60, -\sqrt{34})$$

$$y\text{-int: } x = 0$$

$$y = \sqrt{34} \cos(0 + 0.54) \\ = 5.$$

$$x\text{-int: } y = 0$$

$$\sqrt{34} \cos(x + 0.54) = 0$$

$$x + 0.54 = \frac{\pi}{2}$$

$$x = 1.03$$

$$\text{endpoint } x = \pi$$

$$y = \sqrt{34} \cos(\pi + 0.54) \\ = -5$$

